Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 6: Physical and Chemical Changes

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	6 — Physical and Chemical Changes
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: treating a single energetic/colour/irreversibility clue as decisive, confusing the solid product's mass with conservation of total mass, calling crystallisation or dissolving 'chemical', missing that one event can be physical and chemical at once, mistaking which condition a controlled rust experiment actually isolates, and reading galvanising's sacrificial action as mere physical sealing. Do not write on this paper — mark answers on your answer sheet.

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1. A teacher lists four claims about telling physical from chemical changes. Only one is always true. Which one?

(A) If a change gives out or takes in heat, it must be chemical, because energy only moves during reactions.

(B) If a change cannot be reversed, it must be chemical, because chemical changes are exactly the irreversible ones.

(C) If the original material can be recovered by a purely physical method, no new substance was formed, so the change was physical.

(D) If a colour change is seen, it must be chemical, because the substance has clearly turned into something else.

2. A 6 g strip of magnesium ribbon is burnt completely in open air and the white powder is collected and weighed. The powder weighs more than 6 g. The best explanation is that:

(A) burning created brand-new matter out of the flame's energy, adding mass to the metal.

(B) the dazzling light was absorbed by the powder and stored there as extra weight.

(C) magnesium combined with oxygen from the air, so the solid product (magnesium oxide) carries the original magnesium plus the added oxygen.

(D) mass can never change in any change, so the extra weight is just a weighing error.

3. Three changes are each irreversible by simple means: (i) a clay pot shattering, (ii) a boiled egg setting solid, (iii) a sheet of paper being shredded. Classify them correctly.

(A) all three are chemical, because none of them can be undone by a simple physical step.

(B) (i) physical, (ii) chemical, (iii) physical — only the egg forms a new substance; shattering and shredding leave the same material.

(C) all three are physical, because each only breaks an object into a different form.

(D) (i) chemical, (ii) physical, (iii) chemical — breaking and shredding rearrange bonds while the egg merely changes shape.

4. A hot saturated solution of sugar is prepared to grow sugar-candy (mishri). It is split into two equal portions: portion 1 is left with a clean thread dipped in and cooled slowly over several days; portion 2 is poured into a wide tray kept in a warm, airy place so the water evaporates. Both finally give solid sugar candy. Which statement is correct?

(A) Cooling is physical but evaporation is chemical, because losing water turns the sugar into a new substance.

(B) Both are chemical changes, because solid candy is a different substance from dissolved sugar.

(C) Only evaporation gives real crystals; slow cooling can never bring a dissolved solid out of solution.

(D) Both are physical changes (crystallisation); slow cooling tends to grow larger, purer crystals, while warm-air evaporation removes water faster but can trap more impurities in the solid.

5. Which list is correctly ordered as chemical, physical, chemical?

(A) Melting of butter; rusting of a nail; freezing of cream.

(B) Souring of milk; cooking of rice; dissolving of sugar.

(C) Crystallising sugar; burning of a candle; bending of a wire.

(D) Curdling of milk; sublimation of dry ice; ripening of a banana.

6. A clean iron plate is left in moist air. Each year a little brown rust forms and flakes off, and the bare plate underneath rusts again. After many years the plate is noticeably thinner and lighter, yet all the flakes collected weigh more than the metal that was lost. This is because:

(A) the rust flakes contain trapped air bubbles that make them seem heavier than the iron.

(B) rusting destroys iron and creates rust from nothing, so weights cannot be compared.

(C) the plate must have been impure, since pure iron would gain weight overall instead of thinning.

(D) rust is iron combined with oxygen and water, so the flakes are heavier than the iron consumed, even though the solid plate itself keeps losing iron.

7. Four sealed test tubes each hold an identical iron nail: (P) tap water with air above it, (Q) tap water with all air removed and the tube filled to the top, (R) dry air over a drying agent, (S) tap water with a layer of oil on top. After two weeks, which nail shows the most rust?

(A) P, because it is the only tube giving the nail both liquid water and a fresh supply of air (oxygen).

(B) R, because dry air is rich in oxygen and oxygen is the main thing rusting needs.

(C) Q, because being fully surrounded by water gives the nail the most moisture of all.

(D) S, because the oil layer keeps the nail warm and warmth speeds up rusting.

8. A car body is coated with zinc (galvanised). After a few years a deep scratch reaches the steel, yet the exposed steel at the scratch stays rust-free far longer than bare steel would. The chemistry behind this is best described as:

(A) zinc is more reactive than iron, so it corrodes in preference to the iron and goes on protecting the exposed metal even where the coating is broken.

(B) the zinc flows like a liquid into the scratch and physically seals it shut again.

(C) the scratch is so thin that air and moisture simply cannot fit into it to start rusting.

(D) the zinc has already turned the steel into stainless steel, which no longer rusts at all.

9. A student observes a change and says, 'Heat was given out, so it must be chemical.' Which single counter-observation most directly refutes the rule the student is using?

(A) Burning a strip of magnesium gave out heat and left a white powder that cannot be turned back.

(B) Stirring an anhydrous solid into water warmed the beaker, yet evaporating the water gave back the very same solid unchanged.

(C) Quicklime added to water gave out heat and formed a new substance, slaked lime.

(D) Wood burning in a stove gave out heat and left ash that is not the original wood.

10. Baking soda is heated strongly in a dry test tube. A gas is given off and led through a tube into clear limewater, which slowly turns milky. The milkiness is caused by, and tells us:

(A) tiny gas bubbles scattering light, a purely physical cloudiness with no new substance.

(B) an insoluble white solid (calcium carbonate) forming as carbon dioxide from the heated soda reacts with the limewater, which is a chemical change because a new substance appeared.

(C) water vapour from the soda condensing into a white mist inside the liquid.

(D) the limewater cooling so much that tiny ice crystals form in it.

11. Before storing an iron griddle (tawa) for the rainy season, a cook rubs a thin film of cooking oil over it and says the oil 'keeps it from rusting.' How does the oil work, and what kind of protection is it?

(A) It chemically changes the iron surface into a rust-proof new metal, a chemical method.

(B) It forms a barrier that keeps air and moisture off the iron, a physical method of preventing the chemical change of rusting.

(C) It dries the iron so completely that it can never react again, even after the oil wears away.

(D) It supplies oxygen to the iron in a slow, controlled way so that no harmful rust can form.

12. Two clear solutions are mixed; the mixture stays clear at first, then turns cloudy and a fine solid settles out, and the liquid above is a different colour from either starting solution. The strongest single conclusion is that:

(A) a chemical change occurred, because a new insoluble solid (a precipitate) and a new colour both indicate new substances formed.

(B) a physical change occurred, because mixing two liquids is only a blending and can be undone by stirring.

(C) the solid is just dust that drifted in from the air and happened to settle while the colour faded.

(D) the solid is unreacted starting material that failed to dissolve and slowly fell to the bottom.

13. Clean iron filings are stirred into blue copper sulphate solution and left for an hour. Afterwards a reddish-brown powder has settled and the liquid has turned pale green. Which set of statements is fully correct?

(A) The iron displaced copper from the solution, so reddish-brown copper metal settled out and the liquid is now pale-green iron sulphate; iron did this because it is more reactive than copper.

(B) The iron rusted to the brown powder, and the green is dissolved copper that leaked out of the solution.

(C) The iron simply dissolved like sugar, losing mass, and the green is the old copper sulphate fading in the light.

(D) Nothing chemical happened; the colours are only a trick of the lighting and the powder is undissolved iron.

14. A change shows all of: a sharp new smell, bubbles of gas, AND the original material cannot be recovered by any simple physical step. A careful student still asks for the most decisive reason to call it chemical. The best answer is:

(A) the gas alone proves it, because only chemical changes can ever release a gas.

(B) the new smell alone proves it, because smells are never produced by physical changes.

(C) the inability to recover the original by a physical means shows a new substance formed, and the gas and new smell agree, so it is chemical.

(D) the three clues together cannot decide it; chemistry can only be confirmed by measuring how much heat is released.

15. A sample of common salt is contaminated with powdered charcoal and fine sand, both insoluble in water. To obtain clean salt crystals, the correct order of steps is:

(A) heat the mixture strongly until the charcoal and sand burn off, then keep the salt left behind.

(B) grind it finely and pick the white salt grains out from the dark bits by hand.

(C) dissolve the mixture in water, filter out the charcoal and sand, then evaporate or slowly cool the clear solution so salt crystals form.

(D) leave the mixture in damp air until the sand and charcoal rot away, then scrape off the clean salt.

16. Which single pair correctly classifies BOTH members?

(A) Galvanising an iron sheet (the coating step) — chemical; cooking an egg — physical.

(B) Sublimation of naphthalene balls — chemical; rusting of a gate — physical.

(C) Dissolving washing soda in water — chemical; digesting a meal — physical.

(D) Fermentation of dough by yeast — chemical; sharpening a pencil with a blade — physical.

17. On a gas stove, three things happen at once while water is boiled in a steel pan: the cooking gas (LPG) burns at the burner, the steel pan grows hot, and the water turns to steam. The correct count of physical and chemical changes among these three events is:

(A) all three are chemical, because they all happen because of the burning flame.

(B) all three are physical, because nothing is destroyed — the pan, water and gas are just heated.

(C) two physical (the pan heating and water turning to steam) and one chemical (the LPG burning to new gases).

(D) one physical (the pan heating) and two chemical (the water boiling and the gas burning), because boiling makes a new substance.

18. A steel storage tank will stand outdoors where its protective coating is likely to get knocked and scratched. An engineer compares three options: (i) painting only, (ii) galvanising with zinc, (iii) wrapping in oiled cloth. Which gives the most durable protection once the coating is scratched, and why?

(A) Galvanising, because even at a scratch the more reactive zinc keeps corroding in place of the iron and so protects the exposed steel.

(B) Painting, because paint reacts with the steel to make a permanent rust-proof layer that scratches cannot break.

(C) Oiled cloth, because the oil feeds the steel oxygen slowly so that only harmless rust forms.

(D) All three are equal, since each merely seals the surface and a scratch ruins each one alike.

19. Asked to write the rusting reaction as a word equation, a student writes 'iron + oxygen → rust.' The teacher says it is still incomplete. The most precise correction within Class-7 scope is:

(A) iron + carbon dioxide → rust, because carbon dioxide in the air is what attacks iron.

(B) iron + nitrogen → rust, because nitrogen is the most plentiful gas in air.

(C) iron + oxygen (from air) + water → hydrated iron oxide (rust); both oxygen and water are needed.

(D) iron + salt → rust, because salt is what combines with iron to make the brown solid.

20. After magnesium ribbon is burnt, the white powder is collected. A pinch is shaken with a little water and the liquid is tested with litmus: it turns red litmus blue. The best conclusion is:

(A) the ribbon was only melted and re-solidified, so it kept all its original properties.

(B) the blue colour comes from leftover cleaning grit, not from any new substance.

(C) burning produced a genuinely new substance (magnesium oxide) whose properties differ from the metal, confirming a chemical change.

(D) magnesium metal itself turns litmus blue, so no new substance is needed to explain it.

21. A solid camphor tablet placed in an open dish in a warm room slowly disappears over a week, leaving no liquid and no residue; if a cold plate is held above it, white camphor specks collect on the plate. The disappearing of the tablet and the specks forming on the plate are:

(A) both chemical, because the camphor broke down into a gas and then a new white solid built up.

(B) both physical — sublimation (solid to vapour) followed by deposition (vapour back to solid) of the same camphor.

(C) a chemical change (vaporising) then a physical change (re-forming the solid).

(D) a chemical change in which camphor reacts with air to make a new white powder on the plate.

22. Four single observations are offered, each meant to prove on its own that a chemical change happened. Taken entirely alone, which is the WEAKEST proof of a chemical change?

(A) A gas with a sharp, unfamiliar smell was steadily given off.

(B) A fine white solid suddenly settled out when two clear liquids were mixed.

(C) The material caught fire, giving out bright light and strong heat.

(D) A block of solid was warmed and slowly turned into a clear liquid of the same material.

23. A few drops of dilute hydrochloric acid are put on a piece of marble (limestone); it fizzes, and the gas given off turns limewater milky. From this, the correct conclusions are:

(A) a chemical change occurred and the gas is carbon dioxide, because it turns limewater milky.

(B) a physical change occurred, because the fizzing is only dissolved air escaping like from a cold drink.

(C) the gas is oxygen, because vigorous bubbling shows oxygen is being released.

(D) the gas is hydrogen, because acids always give hydrogen and hydrogen makes bubbles.

24. A stainless-steel water bottle and a chrome-plated bicycle bell both stay shiny and rust-free for years. A student asks why the iron in each does not rust. The best shared explanation is:

(A) in both, the iron has been turned chemically into pure chromium, which cannot rust.

(B) both work only because they are always kept perfectly dry.

(C) in both, a chromium-bearing protective layer or alloy keeps a rust-resistant surface between the iron and the air and moisture.

(D) both speed up the iron's reaction with oxygen so the iron is used up before it can rust.

25. A student claims 'cooking always means a chemical change.' Which kitchen step best shows the claim is too broad because that step is actually physical?

(A) Roasting groundnuts until they brown and smell nutty.

(B) Boiling milk until a skin forms and it can curdle.

(C) Melting a spoon of solid ghee in a warm pan, which sets again as the same ghee when it cools.

(D) Frying a batter until it turns crisp and golden.

26. Four identical iron nails are set up, each changing just one condition: jar 1 — moist air with oxygen; jar 2 — moist air, oxygen removed; jar 3 — dry air with oxygen; jar 4 — dry air, oxygen removed. After a fortnight only jar 1 has heavy rust; the rest are nearly clean. The clearest single conclusion is:

(A) rusting needs BOTH oxygen and moisture together, since heavy rust appears only when both are present.

(B) rusting needs only oxygen, since jar 3 (dry air, oxygen) should then have rusted as much.

(C) rusting needs only moisture, since jar 2 (moist air, no oxygen) should then have rusted as much.

(D) rusting needs neither, since iron rusts in time whatever the conditions.

27. A child says, 'Tinning a steel can and galvanising a steel bucket protect the iron in exactly the same way.' Within Class-7 scope, the most accurate reply is:

(A) they are identical, because both turn the iron into a new rust-proof metal underneath.

(B) tinning makes iron rust faster while galvanising stops it, so they work in opposite ways.

(C) both coat the iron to keep out air and moisture, so the simplest shared idea is that each forms a protective surface layer over the iron, though a break in the coat weakens the protection.

(D) neither tin nor zinc affects rusting; the contents of the can keep the iron safe.

28. Which comparison of rusting, the burning of fuel, and breathing (respiration) is correct?

(A) Only burning is chemical; rusting and respiration are physical because the iron and the body stay the same.

(B) All three are physical, because none makes a new substance.

(C) Rusting and burning are chemical, but respiration is physical because it only gives warmth.

(D) All three are chemical changes in which a substance combines with or uses oxygen — burning is fast, respiration is a slow controlled use inside the body, and rusting is slow surface oxidation.

29. Brine (salt dissolved in water) is split: one half is left in a shallow tray in the sun to dry, the other half is boiled hard in an iron karahi until dry. Both leave white salt. A student says the boiled half 'must be a new substance because so much heat went in.' The correct verdict is:

(A) the boiled half is chemically new, because strong heat always makes a new substance.

(B) both halves give back the same salt; using a lot of heat only to drive off water is still a physical change, since no new substance forms.

(C) the sun-dried half is physical but the boiled half is chemical, because the amount of heat decides the type.

(D) both are chemical, because evaporating water from salt always makes a fresh compound.

30. Which change is the odd one out — the only PHYSICAL change in the list?

(A) Bread dough rising and baking into a loaf.

(B) A cut apple turning brown after a while in the air.

(C) Green grapes drying into raisins by losing water and turning brown and sweet-sour.

(D) Wax solidifying as it drips down the side of a candle and cools.

31. Some pale-green iron sulphate solution is poured into a saucer and forgotten on a windowsill. Over a few days a brown scum forms across its surface. Within Class-7 ideas, the most reasonable explanation is:

(A) the iron in the solution is reacting with oxygen and moisture from the open air to form a brown, rust-like new substance — a chemical change.

(B) the green colour is simply bleaching in the sunlight, a physical change with no new substance.

(C) the solution is crystallising, and the crystals merely happen to look brown.

(D) the iron sulphate is dissolving still further, which always turns a solution brown.

32. Which paired classification is fully correct?

(A) Burning of a candle wick — physical; melting of candle wax — chemical.

(B) Crystallisation of sugar from syrup — chemical; cutting of a wire — chemical.

(C) Digestion of food — physical; ripening of a mango — physical.

(D) Souring of milk into curd — chemical; condensation of steam into water droplets — physical.

33. A student insists, 'Any change that needs heat to make it happen must be chemical.' Which single example best refutes this rule?

(A) Heating sugar in a pan until it turns into brown, bitter caramel.

(B) Heating limestone strongly until it gives off a gas and leaves a new solid.

(C) Heating solid wax until it melts to liquid wax, which sets back to the same wax on cooling.

(D) Heating iron wool in a flame until it glows and turns to dark crumbly bits.

34. A dish of clear limewater is left standing uncovered on a laboratory bench. After a week a thin white film has formed across its surface. The most likely reason within Class-7 chemistry is:

(A) the water slowly evaporated, leaving a physical film of dust on top.

(B) carbon dioxide from the air slowly reacted with the limewater to form a thin layer of insoluble white solid — a chemical change.

(C) the limewater froze a little where its surface was coolest.

(D) oxygen from the air dissolved in and bleached the surface white.

35. When green leaves photosynthesise, carbon dioxide and water are turned into glucose and oxygen using sunlight. A student calls it physical 'because the leaf just gets greener.' The correct reasoning that makes it chemical is:

(A) sunlight is absorbed, and absorbing light is always a chemical change.

(B) new substances (glucose and oxygen) with properties unlike the starting materials are produced, which is the test for a chemical change.

(C) water changes from liquid to vapour inside the leaf, and changing state is chemical.

(D) the process can be reversed by simply placing the plant in the dark, and reversible changes are chemical.

36. A steel screwdriver is stroked many times with a bar magnet until its tip can pick up small screws; later it is dropped hard and heated, and it loses this power. Throughout, the screwdriver stays steel. Magnetising and de-magnetising the screwdriver are:

(A) both chemical changes, because the steel gains and then loses magnetic energy.

(B) a chemical change (magnetising) then a physical change (de-magnetising).

(C) rusting, because heating steel in air always corrodes it into a new substance.

(D) both physical changes, because the screwdriver stays the same steel and no new substance forms.

37. A spoon of sugar is stirred into a glass of plain water and into a glass of water that already has blue copper sulphate dissolved in it. In both it simply dissolves and nothing new is seen. A student claims 'in the copper sulphate glass the two dissolved things must have reacted.' The correct conclusion is:

(A) a reaction occurred in the copper sulphate glass, because two dissolved substances always react.

(B) in both glasses the sugar only dissolved (a physical change); dissolving makes no new substance, so no reaction occurred.

(C) the sugar was destroyed in both glasses, since dissolved sugar can never be got back.

(D) dissolving in copper sulphate water is chemical and in plain water physical, because the colours differ.

38. Identical iron lamp-posts are put up in two towns on the same day: one in a humid town right by the sea, one in a dry desert town inland. After a year the seaside lamp-post is far rustier. Within Class-7 ideas, the best reason is:

(A) seaside air carries more moisture and dissolved salt, and salt speeds rusting, so both moisture and salt accelerate the corrosion.

(B) seaside air is hotter, and heat alone is what turns iron into rust.

(C) seaside air has extra carbon dioxide, which is the gas that rusts iron.

(D) the seaside lamp-post is made of weaker iron that rusts on its own without air or moisture.

39. One change gives out heat as a new substance forms; another takes in heat as a new substance forms. A student is unsure whether the heat-absorbing one can really be chemical. The correct judgement is:

(A) both are chemical, because forming a new substance defines a chemical change whether heat is given out or taken in.

(B) only the heat-releasing one is chemical, because chemical changes must give out heat.

(C) only the heat-absorbing one is chemical, because taking in heat means bonds are breaking.

(D) neither is chemical, because any change involving heat is really physical.

40. A worker heats a straight iron rod until it glows red, bends it into a curved bracket, and lets it cool; ignoring a little surface scale, it is still iron. Two events are (i) bending the hot rod and (ii) the rod glowing red. These are best described as:

(A) both chemical, because heating iron until it glows always turns it into a new substance.

(B) both physical — bending changes only the shape, and glowing is hot iron giving off light, with the iron staying iron in both.

(C) a physical change (bending) and a chemical change (glowing), because the glow proves a reaction.

(D) rusting and crystallisation, because hot iron cooled in air rusts and forms crystals.

41. Which statement comparing crystallisation with rusting is correct?

(A) Crystallisation is a physical change giving back the same substance as pure crystals, while rusting is a chemical change forming a new substance from iron, oxygen and water.

(B) Both are chemical, because each produces a solid that was not there before.

(C) Both are physical, because in each the salt or metal stays the same substance.

(D) Crystallisation is chemical (new crystals form) while rusting is physical (iron stays iron under the rust).

42. A student wants to show that OXYGEN (not just any gas) is needed for rusting, while making sure water is present in both set-ups. The best pair is:

(A) a nail in dry air with a drying agent versus a nail in moist air — comparing two kinds of air.

(B) a nail in water left open to ordinary air versus a nail in water that was boiled to drive out air and then sealed under oil — both wet, differing only in whether air (oxygen) can reach the water.

(C) a nail in salty water versus a nail in plain water — both open to air, comparing two waters.

(D) a painted nail versus a bare nail in damp air — comparing a sealed nail with an open one.

43. A few drops of dark red ink are added to a glass of water, turning it pink; adding more water makes it almost colourless. A student says 'the fading colour proves the ink reacted chemically with the water.' The correct verdict is:

(A) a chemical change occurred, because any colour change proves a new substance.

(B) no chemical change occurred; adding water only spreads the same colouring through more liquid — a physical change recoverable by evaporation.

(C) a chemical change occurred, because water always reacts with a dye to weaken it.

(D) it cannot be decided, because colour changes can never be classed as physical or chemical.

44. A boiled egg cannot be un-boiled, and a shattered glass tumbler cannot be un-shattered. A student concludes 'both are chemical because both are irreversible.' The verdict for the cooking is right, but the reasoning is wrong because:

(A) irreversibility always guarantees a chemical change, so the reasoning is actually fine.

(B) both the cooking and the shattering are in fact physical, so the verdict is wrong.

(C) neither cooking nor shattering forms a new substance, so neither is chemical.

(D) irreversibility alone does not prove a chemical change; cooking is chemical because new substances form, while a shattered tumbler is irreversible yet still the same glass (physical).

45. A change is probed with four checks: (1) an attempt to recover the starting material by a physical method failed; (2) a thermometer showed the temperature rose; (3) no colour change was seen; (4) no gas was detected. Which single result is the strongest evidence the change is chemical?

(A) Check 1: the starting material could not be recovered by a physical method, indicating a new substance formed.

(B) Check 2: the temperature rose, which by itself proves a chemical change.

(C) Check 3: no colour change, which proves the change must be chemical.

(D) Check 4: no gas, which proves a new substance formed.

46. Two changes are compared: (i) an ice cube left on a plate melts into water; (ii) a glass of milk left out for a day turns sour and smells off. Choose the correct classification and the reason that distinguishes them.

(A) both physical, because each is just something left out changing on its own.

(B) both chemical, because both happen due to warmth from the surroundings.

(C) (i) physical, because the ice only changes state to the same water; (ii) chemical, because souring produces new substances and a new smell.

(D) (i) chemical because heat was absorbed, and (ii) physical because the milk merely changed taste.

47. The shiny base of a new copper-bottomed pan is held in a hot flame and turns black on the surface; scraping reveals bright copper underneath. The black coating is best explained as:

(A) soot from the flame merely settling on the copper and physically sticking.

(B) a new substance (a copper-oxygen compound) formed where copper combined with oxygen in the air — a chemical change at the surface.

(C) the copper just getting hot, the black being only the colour of hot copper.

(D) the copper melting and re-freezing black, a physical change of state.

48. Which statement about the energy involved in physical and chemical changes is correct?

(A) Only chemical changes can release energy; physical changes never do.

(B) Only physical changes can absorb energy; chemical changes always release it.

(C) Every energy-releasing change is chemical and every energy-absorbing change is physical.

(D) Both physical and chemical changes can absorb or release energy, so energy flow alone does not tell you which type a change is.

49. Concentrated sugar syrup is spread thin on a metal plate near a warm stove. As it dries, clear sugar crystals appear over most of the plate, but a few brown-black flecks form at the hottest edge where some sugar scorched. The crystal-forming and the scorching are, respectively:

(A) a physical change (crystallisation of the same sugar) and a chemical change (sugar scorching into new substances).

(B) both physical changes, because it is all sugar coming out of the same syrup.

(C) both chemical changes, because the sugar separated from the water in each.

(D) a chemical change (crystallising) and a physical change (scorching), because new crystals are a new substance.

50. A child melts a small lead fishing-sinker on an old spoon, lets it cool to a hard lump, then melts it again — each time the same dull-grey metal. A friend says 'melting a metal must be chemical because metals are special.' The correct response is:

(A) yes, melting a metal is chemical, because heat turns the metal into a new substance each time.

(B) melting and re-solidifying the lead are physical changes of state; the metal keeps its identity, so being a metal does not make melting chemical.

(C) melting is physical but cooling is chemical, because the cooled lump is a new solid.

(D) it is rusting, because heating a metal in air always corrodes it into a new substance.

51. Which one of these everyday events correctly shows BOTH a physical change AND a chemical change happening together?

(A) An ice cube melting in a glass of juice — the ice melting is physical and the juice cooling is chemical.

(B) Sugar dissolving in tea — the sugar spreading is physical and the tea turning sweet is chemical.

(C) A wire being bent — the bending is physical and the wire warming up is chemical.

(D) A candle burning — the solid wax melting is physical, while the wax vapour burning at the flame is chemical.

52. A lab notebook records one experiment: 'When the two were mixed, the beaker grew warm, a pale solid sank to the bottom, a faint new smell rose, a gas bubbled off, and the starting liquid could not be got back by evaporating.' How many of these five observations are recognised clues of a chemical change?

(A) Three, because warmth and the failure to recover by evaporation are not chemical-change clues.

(B) Two, counting only the gas and the solid, since smells and heat are unreliable.

(C) Four, because warmth alone is never a clue to a chemical change.

(D) Five — warmth, a precipitate, a new smell, a gas evolved, and failure to recover the original by a physical step are all clues that a new substance formed.

53. Pure, well-formed copper sulphate crystals are grown by slow cooling from a solution made of a dirty, dull starting powder. The reason crystallisation gives clean crystals is best stated as:

(A) the impurities react with the copper sulphate and are turned into more copper sulphate.

(B) the heat used to dissolve the powder burns all the impurities away.

(C) the crystals are a brand-new compound that no longer contains any of the original impurities.

(D) the soluble copper sulphate dissolves and re-forms as crystals, while insoluble impurities are filtered off and soluble impurities stay behind in the leftover solution.

54. Four iron items are kept for a year with their coatings intact: a spanner smeared with grease and kept in a dry toolbox; a window grille painted and left outdoors; a bare bucket kept in a steamy bathroom; a zinc-galvanised watering can kept outdoors. Rank them from LEAST to MOST rusted.

(A) the greased spanner and galvanised can least, the painted grille next, the bare bucket most — intact grease, zinc and paint all block or sacrificially protect the iron, while the bare bucket in steamy air rusts freely.

(B) the bare bucket least, then the grille, then the can, then the spanner most, because grease attracts moisture and speeds rusting.

(C) all four rust equally, because iron always rusts in a year however it is stored.

(D) the galvanised can most, because zinc reacts with air and rusts faster than bare iron.

55. Two friends argue. Ravi says, 'A change is chemical only if you can see something — a colour, a gas, or a flame.' Meera says, 'Some chemical changes show no obvious sign at the time, yet a new substance has still formed.' Who is correct, and why?

(A) Ravi, because if nothing is visible then no new substance can have formed.

(B) Meera, because the real test is whether a new substance formed, which can be true even when no dramatic sign is seen at the time.

(C) Ravi, because chemical changes are defined as those that give a visible signal.

(D) neither, because a change with no visible sign is always physical by definition.

56. A bright steel wool pad flares up with orange sparks when touched to a flame, leaving dark crumbly bits; another pad left in a damp corner for weeks slowly turns brown and flaky. Both events are best described as:

(A) both chemical changes in which iron combines with oxygen — fast and fiery in the flame, slow as rusting in the damp.

(B) the flame event is chemical but the slow browning is physical, because slow changes are physical.

(C) both physical, because the pad is still made of iron in both cases.

(D) the slow browning is chemical but the fiery burning is physical, because sparks are just hot bits flying off.

57. A student argues, 'Because some changes can be undone and give back the original, reversibility goes with physical changes.' Which single case best SUPPORTS this claim?

(A) Burning paper and then trying to gather the smoke back into paper.

(B) Frying an egg and then cooling it to get the raw egg back.

(C) Dissolving salt in water and then evaporating to recover the very same salt — reversible, with the same substance throughout.

(D) Rusting a nail and then wiping it to turn the rust back into shiny iron.

58. A painted iron car door develops a tiny stone-chip in the paint; a year later a rust blister has spread under the paint around the chip and lifted it. The best explanation is:

(A) the paint itself slowly turned into rust across the whole door at once.

(B) the chip let in paint thinner, which is what corroded the steel.

(C) at the chip, air and moisture reached the bare steel and rust formed; the bulky rust pushed off nearby paint, baring more steel that then rusted further.

(D) the rust blister proves that paint makes steel rust faster than leaving it bare.

59. A controlled test shows: a nail in dry air does not rust; a nail in cooled, freshly boiled water sealed under oil does not rust; a nail in ordinary water open to air does rust. A student concludes 'water by itself causes rust.' The correct conclusion is:

(A) water alone causes rust, since the nail in open water rusted.

(B) air alone causes rust, since the dry-air nail had plenty of air.

(C) water alone is not enough; rusting needs both water and air (oxygen), since the sealed boiled-water nail had water but no air and did not rust.

(D) neither water nor air is needed; the open-water nail rusted only because it was disturbed.

60. A student lists four statements. Which single one is FALSE within Class-7 science?

(A) A chemical change forms one or more new substances, while a physical change does not.

(B) Some physical changes, such as freezing water, release heat to the surroundings.

(C) Crystallisation gives pure crystals of the same substance and is a physical change.
(D) Every change that gives off a gas is a chemical change.

Solutions — Science (Class 7), Chapter 6: Physical and Chemical Changes — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	B	31	A	41	A	51	D
2	C	12	A	22	D	32	D	42	B	52	D
3	B	13	A	23	A	33	C	43	B	53	D
4	D	14	C	24	C	34	B	44	D	54	A
5	D	15	C	25	C	35	B	45	A	55	B
6	D	16	D	26	A	36	D	46	C	56	A
7	A	17	C	27	C	37	B	47	B	57	C
8	A	18	A	28	D	38	A	48	D	58	C
9	B	19	C	29	B	39	A	49	A	59	C
10	B	20	C	30	D	40	B	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The single reliable rule is the new-substance test, and recovering the starting material by a physical means (such as evaporating to get dissolved salt back) proves no new substance ever formed, so the change was physical. The 'heat means chemical' idea fails because dissolving some solids or even melting and freezing exchange heat while staying physical. The 'irreversible means chemical' idea fails because smashing glass is irreversible yet physical. 'Colour change means chemical' fails because heating a wire to glow, or merely diluting a coloured liquid, changes colour with no new substance. Only the recoverability criterion holds in every case.

2. (C) Burning magnesium is a chemical change in which the metal combines with oxygen to make magnesium oxide. The solid product therefore contains the original magnesium atoms plus oxygen atoms taken from the air, so the solid alone weighs more than the ribbon did. Energy (the light and heat) is not matter and does not add weighable mass, so 'matter created from energy' and 'light stored as weight' are both wrong. Total mass counting the oxygen taken from the air is conserved, but that does not stop the solid by itself from gaining mass, so 'mass can never change' misreads conservation as applying to the solid alone.

3. (B) Irreversibility alone does not decide the type; only the formation of a new substance does. The shattered pot is still the same clay material in pieces (physical), and shredded paper is still paper (physical), even though neither can be reassembled. Cooking the egg forms new substances from the egg proteins and cannot be undone (chemical). So the pattern is physical, chemical, physical. 'All chemical because irreversible' and 'all physical because only the form changes' both apply one feature blindly; the bond-rearranging description is exactly backwards, since it is the egg, not the pot or paper, where new substances form.

4. (D) Crystallisation is a physical change: dissolved sugar simply comes out of solution as crystals of the very same substance, whether the solution is cooled (which lowers how much it can hold) or evaporated (which removes the water). Slow cooling usually grows fewer, larger, cleaner crystals, while faster warm-air evaporation can leave impurities behind in the solid. Dissolved sugar and solid sugar candy are the same substance, so it is not chemical, and gentle warm-air drying does not transform sugar into anything new. Both routes yield real crystals, so 'only evaporation works' is false.

5. (D) Curdling forms new acidic substances (chemical); dry ice subliming is solid carbon dioxide turning to gas of the same substance (physical); a banana ripening develops new sugars and aroma compounds (chemical) — so chemical, physical, chemical. 'Melting butter, rusting, freezing cream' is physical, chemical, physical. 'Souring, cooking, dissolving' is chemical, chemical, physical. 'Crystallising sugar, burning candle, bending wire' is physical, chemical, physical. Only the first list matches the required order.

6. (D) Rusting combines iron with oxygen and water to form hydrated iron oxide, so each gram of iron that reacts ends up in a flake that also contains added oxygen and water and therefore weighs more than the iron alone. That is why the collected flakes outweigh the metal lost, while the plate itself thins because iron keeps leaving its surface as rust that crumbles away. Trapped air is far too light to explain the gain; the mass comes from chemically combined oxygen and water. Rust is not made from nothing. Even pure iron thins as a plate while its flakes gain mass, so impurity is not the reason.

7. (A) Rusting needs iron, oxygen, and water together. Tube P has both liquid water and air above it, so it rusts most. Tube R has air but no moisture, so it barely rusts. Tube Q has plenty of water but the air was removed, so it lacks oxygen and rusts very little. Tube S has water but the oil seals out air, so again little oxygen reaches the nail. Dry air alone cannot rust iron because moisture is missing, and merely being surrounded by water without dissolved air does not help, while oil seals air out rather than warming the nail.

8. (A) Galvanising protects partly as a barrier but crucially because zinc is more reactive than iron: at the scratch the surrounding zinc corrodes first and keeps shielding the bare steel, which is why galvanised steel resists rust even where the coating is broken. Solid zinc does not flow to re-seal a scratch. A scratch easily admits air and moisture, so size is not the reason. Galvanising lays a zinc coat over the steel; it does not convert the steel into stainless steel, which would require chromium mixed throughout the metal.

9. (B) The rule 'heat given out means chemical' is broken by a physical change that still gives out heat. Dissolving an anhydrous solid can warm the water, yet evaporating recovers the same solid, proving no new substance formed — a physical change that released heat. Burning magnesium and burning wood do give out heat, but they are chemical (new substances form), so they support the rule rather than refute it. Quicklime plus water also forms a new substance with heat, again chemical. Only the dissolving-and-recovering example shows heat without any new substance, defeating the rule.

10. (B) Heating baking soda drives off carbon dioxide, which reacts with limewater to form insoluble calcium carbonate; its fine white particles make the liquid look milky, and the appearance of a new insoluble solid is a clear chemical-change clue. The clouding is a chemical product, not light-scattering by gas bubbles, which would not persist. Condensing water vapour cannot form a white solid inside the liquid. The breath-free limewater is not freezing; the milkiness appears the moment the carbon dioxide reacts, not from cooling.

11. (B) Oil and grease protect by coating the iron so that air and moisture cannot reach it; the oil itself does not change the iron, so it is a physical barrier method that prevents the chemical change of rusting. It does not turn the iron into a new metal — that would be a chemical change such as galvanising. It does not permanently dry the metal; if the film wears away, air and moisture can reach the iron and rusting begins. Adding oxygen would speed rusting, not prevent it, since rusting needs oxygen.

12. (A) A precipitate forming where the mixture was clear, together with a new colour in the liquid, are two independent clues that new substances were produced — a chemical change. Mixing solutions is not always physical: here a reaction makes an insoluble product, which simple stirring cannot undo. Stray dust would not appear throughout the liquid at the moment of mixing nor change the colour. The solid is not undissolved starting material, because both solutions were clear before mixing; the solid is a product of the reaction between them.

13. (A) Iron is more reactive than copper, so it displaces copper from copper sulphate: copper metal comes out as a reddish-brown powder while iron goes into solution as pale-green iron sulphate. The brown powder is copper, not rust, since rust needs air and water and is a different oxide. The iron does not dissolve unchanged like sugar — it reacts in a displacement; and the green is a genuine new substance, not the old solution fading or a lighting trick.

14. (C) The decisive criterion is whether a new substance formed; failure to recover the starting material by a physical method shows one did, and the new smell and evolved gas are confirming clues of new substances. A gas can also be released physically (boiling water gives steam, opening a fizzy drink lets dissolved gas escape), so a gas alone is not proof. A new smell usually signals new substances but is not by itself the formal test. Measuring heat does not settle it, since physical changes also exchange heat; the recovery test is the reliable one.

15. (C) The salt dissolves while charcoal and sand do not, so dissolving and filtering removes the insoluble dirt; evaporating or cooling the clear filtrate then crystallises pure salt. Strong heating would not cleanly remove sand and charcoal (sand does not burn) and risks spoiling the salt. Hand-picking cannot reliably separate finely mixed powders or give pure crystals. Sand and charcoal do not 'rot' in damp air, so that route fails — only dissolve–filter–crystallise gives clean salt crystals.

16. (D) Yeast fermenting dough produces new substances such as carbon dioxide and alcohol, so it is chemical, while sharpening a pencil only removes shavings, leaving the same wood and graphite, so it is physical — both labelled correctly. Galvanising's coating step lays zinc over iron without changing the iron, so it is physical, and cooking an egg forms new substances, so it is chemical — that pair is reversed. Naphthalene subliming is a state change (physical) and rusting forms a new oxide (chemical) — reversed. Dissolving washing soda is physical (the solid is recoverable) and digestion is chemical — reversed too.

17. (C) The steel pan getting hot and water turning to steam are physical changes — the pan is unchanged and steam is still water, just in vapour form. Only the LPG burning combines fuel with oxygen to make new substances such as carbon dioxide and water vapour — a chemical change. So the count is two physical and one chemical. Being driven by the flame's heat does not make a change chemical (the pan heating is heat-driven yet physical), so 'all chemical' is wrong; the gas does change chemically, so 'all physical' is wrong; and boiling water forms no new substance, so it is not chemical.

18. (A) Galvanising is best for a coating likely to be scratched because zinc is more reactive than iron; at a scratch the zinc corrodes preferentially and goes on protecting the exposed steel, unlike a pure barrier. Paint is only a barrier, not a chemical bond with the steel, and a scratch lets the bare steel rust. Oil also only seals; it does not 'feed oxygen', and feeding oxygen would speed rusting anyway. The three are not equal: paint and oil fail at a scratch, while galvanising's sacrificial zinc keeps working, so a scratch does not ruin all three equally.

19. (C) Rusting requires iron to combine with oxygen from the air and with water (moisture) to form hydrated iron oxide, so 'iron + oxygen' is incomplete because it leaves out the essential water. Carbon dioxide is not the reactant in rusting — that gas is used in the limewater test, not corrosion. Nitrogen, though most plentiful in air, does not rust iron.

Salt can speed up rusting but is not consumed into the rust and cannot replace the essential water, so 'iron + salt' is wrong.

20. (C) The white powder dissolving to give a mildly basic solution shows it has properties the shiny metal ribbon never had, which is direct evidence that burning made a new substance (magnesium oxide) — a chemical change. The ribbon was not merely melted and re-solidified; it was chemically converted, so it does not keep the metal's properties. The basic behaviour comes from the oxide product, not from sandpaper grit used to clean the ribbon. Magnesium metal does not turn litmus blue by itself, so a new substance is indeed needed to explain the result.

21. (B) Camphor sublimes directly from solid to vapour when warmed, and the vapour deposits back to solid camphor on the cool surface — both are physical state changes of the very same substance, with no new substance formed (no liquid stage is needed). Heating does not break camphor into new substances and rebuild it, so 'both chemical' is wrong. Vaporising is not chemical; the substance stays camphor, so the mixed labels are wrong. Camphor does not react with air to make a new powder; the white specks on the plate are the original camphor.

22. (D) A solid melting into a liquid of the same material is simply a change of state and is almost always physical, so by itself it is the weakest evidence of chemistry. A new-smelling gas suggests a new substance, a precipitate from mixing two solutions is a recognised chemical clue, and catching fire with light and heat points to combustion — all are far stronger indicators of a chemical change than a mere change of state.

23. (A) The acid reacting with the marble makes new substances and releases a gas that turns limewater milky; only carbon dioxide gives that milky test, so the gas is carbon dioxide and the change is chemical. The fizzing is not dissolved air escaping as in a fizzy drink (which is physical); here a reaction generates new gas. The gas is not oxygen — oxygen does not turn limewater milky. Within this Class-7 limewater test the milkiness identifies carbon dioxide, not hydrogen, so 'hydrogen' is wrong.

24. (C) Chrome plating coats the iron with a thin rust-resistant chromium layer, and stainless steel is an iron alloy containing chromium that gives a corrosion-resistant surface — in both, a chromium-bearing surface keeps air and moisture from attacking the iron beneath. The iron is not transmuted into pure chromium; it stays iron with a protective surface. They resist rust even when wet (a bottle is washed daily), so 'only if kept dry' is wrong. They slow the iron's reaction with oxygen rather than speeding it up, so the last option is the opposite of the truth.

25. (C) Melting ghee only changes its state, and it re-solidifies as the same ghee on cooling, so it is physical — showing not every kitchen step is chemical. Roasting groundnuts browns them by forming new substances with a nutty smell (chemical), boiling milk until a skin forms and it can curdle forms new substances (chemical), and frying batter to a crisp golden crust forms new substances (chemical). All three of those are genuine chemical changes, so only melting ghee refutes the over-broad claim.

26. (A) Only jar 1, which has both oxygen and moisture, rusts heavily; removing either oxygen (jar 2) or moisture (jar 3), or both (jar 4), almost stops rusting. That pattern shows both factors are needed together. 'Only oxygen' is wrong because jar 3 had oxygen in dry air yet barely rusted. 'Only moisture' is wrong because jar 2 had moisture without oxygen yet barely rusted. 'Neither' is wrong because jar 1 rusted heavily precisely when both were present.

27. (C) Both tin and zinc are applied as protective surface coatings that keep air and moisture off the iron, which is the core Class-7 idea about coating metals to prevent rusting; in both cases an intact coat protects and a break in the coat lets attack begin. Neither turns the iron into a new metal beneath; the iron stays iron. Tin does not make iron rust faster — it is a protective coat, not an accelerator. The protection comes from the metal coatings, not from the can's contents, so the last option is wrong.

28. (D) Rusting, burning, and respiration all involve oxygen and all form new substances: burning oxidises a fuel rapidly with heat and light, respiration uses oxygen to break down sugar inside the body releasing energy, and rusting is slow oxidation of iron at its surface — so all three are chemical. They are not physical: each makes new substances. Respiration is not merely physical warmth; it breaks sugar into carbon dioxide and water, so the options calling some of them physical are wrong.

29. (B) Evaporating the water — whether slowly in the sun or quickly over a fierce flame — only removes the water and leaves the same salt behind, so both are physical changes; the amount of heat used does not change a physical separation into a chemical one. Strong heat does not automatically create a new substance; here it merely drives off water faster. The salt is recoverable and unchanged in both halves, so neither half is chemical, and evaporating water from salt does not make a new compound.

30. (D) Liquid candle wax cooling and solidifying is just a change of state of the same wax, so it is physical and the odd one out. Dough baking into bread forms new substances (chemical). A cut apple browning in air is a chemical change as it reacts with oxygen, forming new brown substances. Grapes drying into raisins lose water but also develop new substances as they age and change flavour and colour, which is treated as a chemical change here rather than mere drying; the safest single physical example is the solidifying wax.

31. (A) Iron sulphate exposed to open air can react with oxygen and moisture to form a brownish iron-oxide-type substance at the surface — a chemical change producing a new substance, similar to the chemistry of rusting. The green is not merely bleaching in light; a new brown substance appears. Crystallisation would give green crystals of the same iron sulphate, not a brown surface scum. Dissolving further would not turn the solution brown; the brown colour signals a new substance from reaction with air.

32. (D) Milk souring into curd forms new acidic substances (chemical), and steam condensing into water droplets is a change of state of the same water (physical) — both labelled correctly. A burning wick makes new substances (chemical) and melting wax is a state change (physical), so that pair is reversed. Crystallising sugar is physical and cutting a wire is physical, so calling them chemical is wrong. Digestion is chemical and mango ripening is chemical, so calling them physical is wrong.

33. (C) Melting wax needs heat yet only changes the state of the same wax, which sets back to the same wax on cooling — a heat-driven physical change that refutes the rule that needing heat makes a change chemical. Caramelising sugar forms new substances (chemical), heating limestone gives off a gas and leaves a new solid (chemical), and burning iron wool makes a new iron-oxygen substance (chemical). All three of those are genuinely chemical, so only the melting of wax shows that needing heat does not make a change chemical.

34. (B) Air contains a little carbon dioxide, which reacts even slowly with limewater to form insoluble calcium carbonate, appearing as a faint white film — the same chemical reaction as the limewater test, just slower. Evaporation would leave behind dissolved lime, not a fresh white insoluble layer formed by reaction. Limewater does not freeze at ordinary room temperature. Oxygen dissolving does not bleach or whiten limewater; the white solid is the carbonate product of reaction with carbon dioxide.

35. (B) Photosynthesis builds glucose and releases oxygen from carbon dioxide and water — these products have different properties from the reactants, and forming new substances is exactly what makes a change chemical, regardless of any shade change in the leaf. Merely absorbing sunlight is not in itself a chemical change. Water turning to vapour is a physical state change, not the reason photosynthesis is chemical. Placing the plant in the dark does not 'reverse' photosynthesis (respiration is a separate reaction), and reversibility is not the test for being chemical.

36. (D) Gaining or losing magnetism changes only a physical property of the screwdriver; it stays the same steel with no new substance, so both magnetising and de-magnetising are physical changes. Magnetic 'energy' is not a substance, so neither step is chemical. There is no new substance in either, so the mixed-label option is wrong. Brief heating to de-magnetise is not the same as rusting, which needs moisture and air over time and forms a new oxide; here the screwdriver stays steel.

37. (B) Dissolving sugar in either glass is a physical change: the sugar spreads through the water but stays sugar and can be recovered by evaporation, and no new substance or visible reaction appears in either. Two dissolved substances do not 'always react'; here nothing reacts. The sugar is not destroyed — it can be recovered by evaporating the water.

The presence of colour from copper sulphate does not make the dissolving chemical; the sugar simply dissolves in both, so both are physical.

38. (A) Coastal air is humid and salty; moisture is one of the two essentials for rusting, and dissolved salt speeds the process, so a lamp-post near the sea rusts faster. Heat alone does not rust iron — without moisture and oxygen there is no rust. Carbon dioxide is not the gas responsible for rusting; oxygen is. The seaside lamp-post is not a 'weaker iron that rusts without air or moisture'; rusting always needs oxygen and water, and the sea simply supplies more moisture and salt.

39. (A) A chemical change is defined by forming a new substance, and chemical changes can either release or absorb heat. The first change (heat out, new substance) and the second (heat in, new substance) both form new substances, so both are chemical regardless of the heat direction. It is not true that a chemical change must release heat, nor that only heat-absorbing changes are chemical, nor that any heat involvement makes a change physical.

40. (B) Bending the red-hot rod only changes its shape, and a red glow is just very hot iron radiating light; in both events the metal stays iron with no new substance, so both are physical. Heating iron until it glows does not turn it into a new substance, so 'both chemical' is wrong. The glow is incandescence, not proof of a reaction, so the mixed label is wrong. Quick shaping and cooling is not rusting (which needs time, moisture and air) nor crystallisation (which means crystals coming out of a solution).

41. (A) Crystallisation separates a dissolved solid out as crystals of the very same substance, so it is physical, whereas rusting combines iron with oxygen and water to make a new substance, hydrated iron oxide, so it is chemical — the two differ in whether a new substance forms. They are not both chemical: crystallisation makes no new substance. They are not both physical: rusting does. The last option reverses the truth, since crystallisation is physical and rusting is chemical.

42. (B) To isolate oxygen while keeping water present, both set-ups must be wet and differ only in air access: a nail in water open to air (oxygen available) versus a nail in boiled water sealed under oil (air expelled) does exactly that, and only the open one rusts, showing oxygen is needed. The dry-versus-moist pair tests moisture, not oxygen, and is not kept wet in both. Salty versus plain water tests salt, with oxygen present in both. Painted versus bare changes the barrier, not cleanly the oxygen, so it does not isolate oxygen.

43. (B) Adding water only disperses the same colouring substance through a larger volume, so the colour looks paler; the dye is unchanged and could be recovered by evaporating the water, making this a physical change. A colour change is not automatic proof of chemistry — here it is just dilution. Water does not react with the dye to weaken it;

it merely dilutes. It is decidable: the recoverability test shows no new substance, so it is physical, not undecidable.

44. (D) Cooking the egg is chemical because new substances form from the proteins, but a shattered glass tumbler is also irreversible while staying the same glass material — a physical change — which shows irreversibility by itself does not prove chemistry. So the cooking verdict is correct but the reasoning (irreversible therefore chemical) is flawed. Irreversibility does not guarantee chemistry. The cooking is genuinely chemical, so 'both physical' is wrong, and cooking does form new substances, so 'neither forms a new substance' is wrong.

45. (A) Failing to recover the original by any physical means is strong evidence that a new substance formed, which is the defining test for a chemical change. A temperature rise alone is not proof, since some physical changes (such as dissolving certain solids) also release heat. The absence of a colour change does not prove chemistry — many chemical changes show no colour change and many physical ones do. The absence of a gas likewise proves nothing about new substances. So only the recoverability result is decisive here.

46. (C) Melting ice is a state change of the same water (physical), while milk going sour involves microbes producing new acidic substances and a new smell (chemical), so the distinguishing test is whether new substances appear. Being 'left out in the air' does not make a change physical or chemical by itself, so 'both physical' is wrong. Warmth alone does not make a change chemical, so 'both chemical' is wrong. Absorbing heat does not make melting chemical, and souring forms new substances, so the last option is reversed.

47. (B) When copper is heated in air, its surface combines with oxygen to form a black copper-oxygen compound — a new substance, so it is a chemical change, which is why fresh copper lies beneath the black layer. Soot is loose flame deposit, but here a layer forms by reaction with oxygen on clean copper even in a clean flame. Hot copper glows but does not stay black after cooling unless a new substance formed. The pan base was not melted and re-frozen; the black is a chemical product, not a change of state.

48. (D) Energy can flow either way in both kinds of change: melting absorbs heat and freezing releases it (physical), while burning releases heat and some reactions absorb it (chemical), so energy flow by itself does not reveal whether a change is physical or chemical. It is false that only chemical changes release energy (freezing releases heat). It is false that chemical changes always release energy or that physical changes never absorb it. Sorting changes by energy direction does not match the physical/chemical divide, so the last option is wrong.

49. (A) As the water evaporates the dissolved sugar comes out as crystals of the very same sugar, which is crystallisation, a physical change. Where the sugar dried and scorched at the hot edge it browned into a new substance, a chemical change. So one event is physical and the other chemical. It is not 'both physical', because the scorching makes a new substance. It is not 'both chemical', because crystals of the same sugar are not a new substance. The last option reverses the labels — crystallising is the physical one and scorching the chemical one.

50. (B) Melting lead and letting it solidify are changes of state of the same metal, which keeps its identity and can be re-melted unchanged, so they are physical changes; being a metal does not turn melting into a chemical change. Heat does not change the lead into a new substance, so 'melting a metal is chemical' is wrong. Cooling back to a lump gives the same metal, so it is not a new solid. Briefly melting on a spoon is not rusting, which needs moisture and time and forms a new oxide.

51. (D) A burning candle famously shows both: solid wax melting to liquid is a physical state change, while the wax vapour combining with oxygen at the flame to form new gases and soot is a chemical change. The other options pair a physical event with something that is also physical: juice getting colder is just heat moving (physical), tea turning sweet is still just dissolved sugar (physical), and a wire warming from bending is heat from friction, not a new substance. Only the candle has a true chemical change alongside the physical one.

52. (D) Warmth (heat released), a precipitate, a new smell, an evolved gas, and the inability to recover the original by a physical means are all recognised clues that point to a chemical change here — that is five. While heat or a gas alone might be ambiguous, in this combined set each item is acting as a chemical-change clue, and the recovery failure clinches that a new substance formed. So the count is not three, two, or four; all five observations are clues in this case.

53. (D) Crystallisation purifies physically: copper sulphate dissolves and is separated from insoluble dirt by filtering, then re-forms as crystals on cooling, while soluble impurities remain dissolved in the mother liquor — so the crystals are the same copper sulphate but cleaner. The impurities are not chemically converted into copper sulphate. Heating to dissolve does not burn impurities away. The crystals are the same compound as the starting copper sulphate, not a brand-new substance, so the purity comes from separation, not from making something new.

54. (A) Intact coatings prevent rusting: grease blocks air and moisture, zinc both blocks and sacrificially protects, and paint blocks air and moisture, while the bare bucket in steamy air has both oxygen and moisture and rusts the most. So the bare bucket is worst and the protected items are far better. Grease does not attract moisture to speed rusting; it repels it. Iron does not rust equally regardless of storage — coatings make a large difference. Zinc protects iron rather than rusting faster than bare iron, so the galvanised can is not the most rusted.

55. (B) Meera is correct: the defining feature of a chemical change is the formation of a new substance, and that can occur even when the visible clues (colour, gas, flame, precipitate) are faint or absent at the moment of observation. Ravi is wrong that no visible clue means no new substance — clues are helpful signs, not the definition. Chemical changes are not defined by producing a visible signal. A change with no obvious clue is not automatically physical; one must test whether a new substance formed, so 'neither' is wrong.

56. (A) Burning steel wool in a flame and rusting iron in damp air both have iron combining with oxygen to form new iron-oxygen substances, so both are chemical changes — one rapid, one slow. Speed does not decide the type: a slow change can be chemical, so 'slow changes are physical' is wrong. Both form new substances, so 'both physical' is wrong. Fiery burning is not physical merely because sparks fly; the sparks are bits of iron reacting with oxygen, so the last option misclassifies the burning.

57. (C) Dissolving salt and recovering it by evaporation is reversible and keeps the same substance throughout, supporting that this reversible change is physical. Burning paper cannot be reversed by gathering smoke — it is chemical and irreversible. Frying an egg and cooling it does not give back the raw egg — cooking is chemical and irreversible. Wiping a rusted nail does not turn rust back into shiny iron — rusting is chemical and not undone by wiping. Only the salt example is a reversible physical change.

58. (C) Paint protects only as a barrier; a chip lets air and moisture reach the bare steel, which rusts, and because rust is bulky and flaky it lifts the surrounding paint, baring more steel that then rusts too — so the spot spreads. The paint does not itself turn into rust; only the steel does. There is no 'paint thinner' attacking the steel through the chip. Paint slows rusting compared with bare steel; the spreading spot is due to the breach, not because paint speeds rusting, so the last option is wrong.

59. (C) The boiled-water-sealed nail had water but no dissolved air and did not rust, which shows water by itself is not enough — air (oxygen) is also required. The open-water nail rusted because it had both water and air. The dry-air nail had air but no moisture and did not rust, so air alone is not enough either. Both water and air are needed together. 'Neither needed' is contradicted by the open-water nail rusting precisely when both were present, not from being disturbed.

60. (D) The false idea is that producing a gas always means a chemical change: boiling water produces steam (a gas) and opening a fizzy drink lets dissolved gas escape, both physical, so gas evolution is only sometimes a chemical clue. The other three are true: a chemical change forms new substances while a physical change does not; freezing is a physical change that releases heat; and crystallisation yields pure crystals of the same substance and is physical.